



FREELANCE SENIOR FULL-STACK & AI ENGINEER

Nikita Petrich

"Anything done manually twice gets automated"

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LOCATION: Remote (preferred) · Munich: 1–2 days/week · Travel: 1–2 days/month

ONBOARDING: On-site at the start (if needed)

AVAILABLE: immediately · full time **RATE:** on request

QUALIFICATION: IHK-certified IT specialist (Cologne) **EXPERIENCE:** 7+ years

FOCUS

LLM integration RAG AI-assisted automation

Agentic Coding / AI-Engineering TypeScript Python

Backend (NestJS · Node.js · FastAPI) Frontend (Next.js · React · Angular)

PostgreSQL Clean Architecture Microservices CI/CD Docker

GDPR-compliant AI architecture

LANGUAGES

German · Native English · Business English · B2 certified · US engagement

WAYS OF WORKING

results-driven self-directed & dependable

autonomous, asynchronous working style clear communication (DE/EN)

PROFILES

[Website](#) [↗]

[LinkedIn](#) [↗]

[GitHub](#) [↗]

[freelancemap](#) [↗]

[Malt](#) [↗]

[Certificates](#) [↗]

i I am a **freelance Senior Full-Stack & AI Engineer** focused on **AI: LLM integration, RAG and AI-assisted automation** – from architecture through to **stable production operation**.

My guiding principle: anything done manually twice gets automated. That applies to my clients' processes just as much as to my own work. In **logistics**, a route management system I built saves up to **1,000 hours per employee per year** through automated document processes; in **retail**, up to **40 hours of administrative work per month**; at a **notary's office**, clients' phone enquiries drop by up to **70 %**.

My own development follows the same principle: I work with **modern agentic coding and AI engineering workflows** – **spec-driven, AI-assisted and with automated tests and reviews**. That gives me **high delivery speed at consistent quality**, with no compromise on **maintainability, traceability and stability**.

I work **independently, remotely and asynchronously**. A clear goal is enough – I structure the path there myself. **I document decisions, code arrives tested and ready for review**, and I raise technical risks early. Communication in **German and English**.

More than **7 years of experience** in **LegalTech, HealthTech, e-commerce, EdTech and logistics** – both as part of a **20-strong engineering team** on an **AI-assisted LegalTech platform with more than 3,000 customers** and as the **sole developer responsible for architecture, operations and further development**.

Does this fit what you have in mind? Book a [free intro call](#) [↗]

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LEGALTECH / GOVTECH

04/2025 – PRESENT · 16 MONTHS

01

BESCHEIDKLAR — AI-Assisted LegalTech SaaS Platform

Role: Founder & CTO Client: [Own product](#) ↗



ROLE Founder & CTO	TEAM Solo · own product (side project)	LOCATION Lower Saxony, Germany · Remote
LANGUAGE German	WEBSITE bescheidklar.de ↗	METHODOLOGY Direct product ownership

RESPONSIBILITIES

- Built a modular, service-oriented architecture for the claimant-facing frontend, the law firm area and the regional licensing model (**Stripe**).
- **AI pre-assessment pipeline** for official notices and terminations (in progress): extraction → deadline check → plausibility and prospects assessment, with the case prepared in structured form.
- Automated, **GDPR-compliant** handover of qualified leads including uploaded documents to matching specialists (lawyers, law clinics).
- **RAG-based** guidance section with articles and recommended next steps for those affected.
- Coverage of several legal areas (civil, social and employment law) through an extensible, multilingual and low-barrier web interface.
- End-to-end responsibility for the tech stack, hosting on EU infrastructure and a secure-by-design compliance concept (**EU AI Act**, **GDPR**, **German Legal Services Act**).
- Type-safe end-to-end data flows (**TypeScript**, **DTOs**, **Zod**) with automated **CI/CD** delivery (**GitHub Actions**, **Docker**), monitoring and error tracking (**Datadog**) and test coverage (**Vitest**, **Jest**, **Playwright**).
- Close collaboration in code reviews and use of AI-assisted tools (**Claude Code**, **Cursor AI**, **Code Rabbit**) to safeguard quality and development pace.

BESCHIEDKLAR — AI-ASSISTED LEGALTECH SAAS PLATFORM — CONTINUED

RESULTS

- **MVP** delivered single-handedly — architecture, platform and operation on EU infrastructure from one source. The remaining scope is in progress.
- Compliance built into the product design rather than retrofitted: the **EU AI Act**, the GDPR and the limits of the **German Legal Services Act** were part of the plan from the start.
- Delivery fully automated: tests, build and container deployment run through **GitHub Actions**, and production errors surface through **Datadog** — releases with no manual step, even as a one-person team.
- Outlook: up to 10,000 automated pre-assessments per month and up to **80 %** less manual case intake.
- Outlook: a predictable inflow of pre-qualified cases for partner law firms instead of costly manual initial assessment.

TECHNOLOGIES

TypeScript JavaScript Python SQL Bash HTML CSS JSON YAML UML Next.js React
 Server-Side Rendering (SSR) FastAPI Backend development Frontend development Web development
 Artificial intelligence GenAI AI Engineering LLM integration OpenAI API Azure OpenAI
 RAG (retrieval augmented generation) Prompt Engineering Embeddings Vector databases pgvector
 Semantic search OCR Document analysis LangChain AI agents Multi-agent workflows
 AI-assisted automation AI-assisted test generation GDPR-compliant AI architecture Claude Code
 Claude Code Hooks Slash Commands Agent Skills Subagents ChatGPT v0 Agentic Coding
 Agentic Software Engineering Context Engineering Spec-Driven Development PostgreSQL Redis
 Data architecture Tailwind CSS shadcn/ui Component libraries Storybook Design Tokens Wireframing
 Internationalisation (i18n) Accessibility (WCAG) German Accessibility Reinforcement Act (BFSG)
 European Accessibility Act (EAA) Stripe Payment processing SEPA JWT OAuth 2.0 Secure by Design
 IT security End-to-end encryption Docker GitHub Actions CI/CD Continuous Integration DevOps Hetzner
 IONOS EU hosting Linux Ubuntu Datadog Monitoring Observability Vitest Jest Playwright
 Unit Testing Integration tests End-to-end tests API testing Postman Test automation Zod DTOs Slack
 Husky Software architecture API-First Middleware Design Patterns Domain-Driven Design Microservices
 Scalable architecture Clean Code SOLID Clean Architecture MVP development SaaS Law firm software
 Process digitalisation EU AI Act AI Governance GDPR Privacy by Design Data minimisation
 Data processing agreement (DPA) Digital sovereignty Kanban Project management IT strategy Digital strategy
 Digital transformation Software development

LEGALTECH / IMMIGRATION

12/2025 – 04/2026 · 5 MONTHS

02

Manifest OS — AI-Assisted Immigration Platform

Role: Senior Full-Stack Engineer (AI feature integration) Client: [Manifest Law](#) ↗

↗ Testimonial



ROLE Senior Full-Stack Engineer (AI feature integration)	TEAM ~20 engineers across 4 teams · platform, AI, full-stack	LOCATION New York, USA · Remote
LANGUAGE English · EU & US	WEBSITE manifestlaw.com ↗	METHODOLOGY Scrum · autonomous · with PM and design

RESPONSIBILITIES

- Production **AI enrichment** of initial legal consultation leads (**HubSpot**, **Cal.com**, **Bluedot**): recordings are transcribed, passed to the OpenAI integration through a **webhook** and a **BullMQ** queue, and written back into the core systems in structured form — the basis for more efficient case handling.
- **AI-assisted evidence processing**: uploaded case documents are analysed, classified and consistently named automatically.
- Consolidation of four portals (**B2B**, **B2C**, lawyer, ops) into a single "**B2X**" portal — in a team of four engineers, focusing on the full-stack migration of the **B2B** and ops areas.
- Introduction of application-wide event tracking (**Amplitude**) as the data foundation for product and data analytics.
- Resilience of the **event-driven** services (**NestJS**, **BullMQ**): retry strategies, idempotency and defined behaviour when the AI and transcription services fail.
- Delivery within a monorepo with clearly scoped feature modules, so it can be split into **microservices** quickly later on, with type-safe interfaces (**TypeScript**, **DTOs**) and automated tests (**Jest**, **Playwright**) in the **CI/CD** pipeline.
- Close collaboration in code reviews and use of AI-assisted tools (**Claude Code**, **Cursor AI**, **Code Rabbit**) to safeguard quality and development pace.

MANIFEST OS — AI-ASSISTED IMMIGRATION PLATFORM — CONTINUED

RESULTS

- AI features running in production: consultation leads are enriched without manual rework — summary, next steps and ownership sit right on the case in the system.
- Application-wide event tracking established as a shared data foundation for product and data analytics.
- Refactoring initiatives in the migrated **B2B** and ops areas: existing structures simplified and geared towards better maintainability.
- According to the provider: more than 3,000 clients, 150+ corporate programmes and up to **15 %** higher approval rates than the USCIS average (a team result).

TECHNOLOGIES

TypeScriptJavaScriptSQLHTMLCSSJSONYAMLNext.jsReactServer-Side Rendering (SSR)

NestJSNode.jsFastifyExpress.jsDependency InjectionREST APIWebhooksDTOsBullMQ

RabbitMQCachingBackend developmentFrontend developmentWeb developmentArtificial intelligence

GenAIAI EngineeringLLM integrationOpenAI APIAnthropic Claude APIModel Context Protocol (MCP)

BluedotHubSpotCRMCal.comAmplitudeProduct AnalyticsPostgreSQLRedisMikroORM

Drizzle ORMTanStack Querybetter-authJWT DockerLinuxUbuntuGoogle Cloud Platform (GCP)

VercelGitHub ActionsCI/CDContinuous IntegrationDatadogMonitoringObservabilityJestPlaywright

End-to-end testsClaude CodeCursor AICode RabbitAgentic CodingAgentic pull request workflows

Code reviewsLinearFigmaSlackHuskyGoogle CalendarGoogle Workspace (Meet, APIs)Monorepo

TurborepoMicroservicesDomain-Driven DesignScalable architectureEvent-Driven ArchitectureIdempotency

Clean ArchitectureAgile methodsDigital transformationSoftware developmentInterface development

API integrationThird-party system integrationRefactoringLegacy modernisationLegalTechScrum

SOCIAL MEDIA / MOBILE APP

11/2025 – 12/2025 · 2 MONTHS

03

RateUp — Rating App for Web, iOS and Android**Role:** Sole Developer · Full-Stack **Client:** [Krutzfeld Tech GmbH](#) ↗

ROLE Sole Developer · Full-Stack	TEAM Solo · client project	LOCATION Zurich, Switzerland · Remote
LANGUAGE German	WEBSITE krutzfeld.tech ↗	METHODOLOGY Self-directed

RESPONSIBILITIES

- Social media rating app from a single **ionic/Angular** codebase — on the web and as native builds for iOS and Android.
- Sign-up, login and onboarding including profile creation (**Firestore Authentication**), secured through **Firestore security rules**.
- Social media feed with filters, own posts and media upload (**Cloud Firestore, Cloud Storage**).
- Delivery of the web version through **Firestore** Hosting with a **CI/CD** pipeline (**GitHub Actions**).
- A maintainable codebase for handover: clearly layered structure, reusable components, type-safe models and a consistent clean-code style (**SOLID**).

RATEUP — RATING APP FOR WEB, IOS AND ANDROID — CONTINUED**RESULTS**

- Finished product handed over to the client — running as a cross-platform app on the web and as a native app on iOS and Android from one shared codebase.
- Built entirely on **Firebase**: authentication, data storage, media storage and delivery — access runs straight through the **Firebase** SDK, with no server layer of its own.
- A codebase ready for handover: consistent structure, reusable components and type safety throughout — the client's team can carry on without a ramp-up hurdle.
- One revision, three platforms: web deployment through **Firebase** Hosting, iOS and Android builds from the same codebase.
- Built on an existing clickable prototype and taken to a product ready for handover in around two months.

TECHNOLOGIES

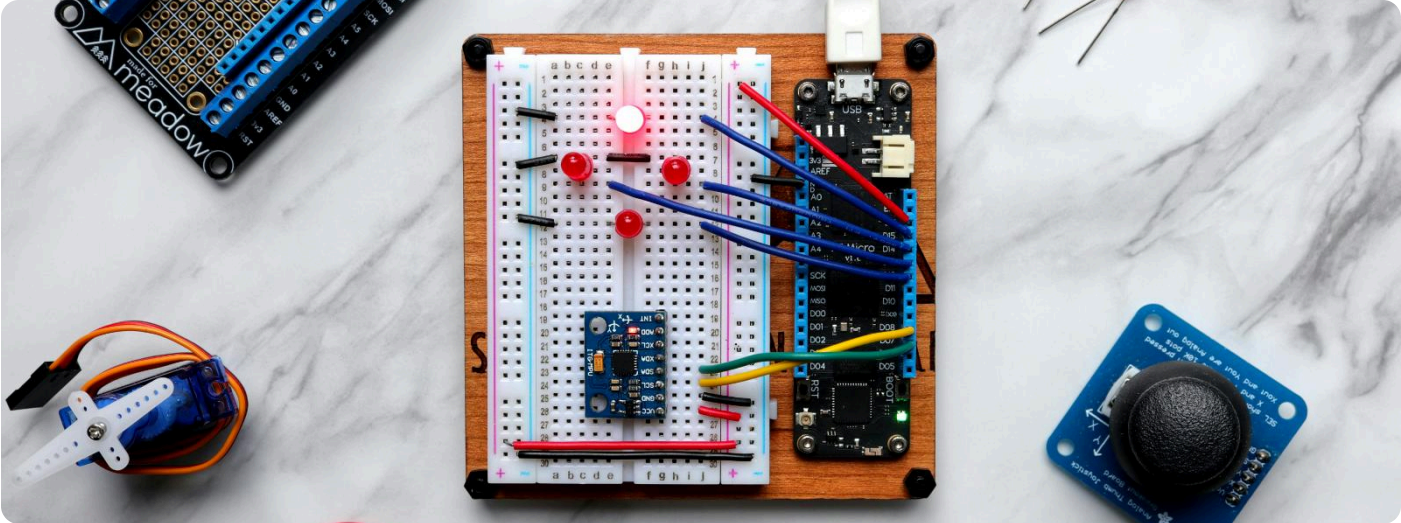
TypeScript JavaScript HTML CSS JSON YAML Angular Ionic RxJS Component libraries
 Cross-platform development iOS Android Mobile First Responsive Design UI implementation User guidance
 Onboarding flows Frontend development Web development Firebase Firebase Authentication
 Firebase Security Rules Cloud Firestore NoSQL Data modelling Cloud Storage Firebase Hosting
 Authentication IT security Social Media Type safety Clean Code SOLID Separation of Concerns
 Clean Architecture Domain-Driven Design Scalable architecture MVP development Git GitHub Trello
 GitHub Actions CI/CD Continuous Integration Linux Digital transformation Software development
 Interface development API integration

IOT / CONSUMER ELECTRONICS

08/2025 – 09/2025 · 2 MONTHS

04 AITOI — Interactive IoT Toy (MVP)

Role: Frontend Engineer Client: AITOI [Testimonial](#)



ROLE Frontend Engineer	TEAM 3 people · frontend, embedded/hardware, lead	LOCATION Frankfurt, Germany · Remote
LANGUAGE German & English	VISIBILITY Internal system	METHODOLOGY Self-directed

RESPONSIBILITIES

- Setup app for commissioning new devices: Wi-Fi onboarding by QR code scan with a guided flow, device pairing and robust error handling.
- PWA dashboard for parents: follow, filter and review their children's play sessions.
- Later reconfiguration of paired devices through the same platform-independent PWA.
- Integration of **Supabase** for authentication, data storage and real-time synchronisation (including **row level security**).
- A fast PWA architecture (**Next.js**) with a responsive UI.
- CI pipeline (**GitHub Actions**) on top of containerised builds (**Docker**): unit tests (**Vitest**) and the build ran automatically on every change; deployment was done by hand.
- Definition and implementation of type-safe data models and API interfaces (**TypeScript**, **DTOs**, **Zod**).
- Close coordination in English with the hardware team to define the device-app communication and the setup protocols.
- Advising and guiding the client from the initial idea to an actionable **MVP** concept.

AITOI — INTERACTIVE IOT TOY (MVP) — CONTINUED

RESULTS

- A fully functional **MVP** frontend in ~2–3 weeks of actual working time — independently, from concept through client coordination to the demo.
- Setup app and parent dashboard as one platform-independent PWA, synchronised with the IoT device in real time.
- A scalable, documented architecture as a **solid** basis for the product launch.
- Trusted, cross-functional collaboration with the hardware team throughout the project — no friction at the device-app interface despite the tight schedule.
- The client's testimonial explicitly names the **CI/CD** pipelines and unit tests as contributing to the quality, security and reliability of the applications.

TECHNOLOGIES

TypeScriptJavaScriptHTMLCSSSQLJSONYAMLNext.jsReactServer-Side Rendering (SSR)

Progressive Web App (PWA)Mobile FirstResponsive DesignDesign SystemComponent librariesUI/UX Design

UsabilityPrototypingReact Hook FormDTOsFrontend developmentBackend developmentWeb development

Device PairingSupabaseSupabase RealtimeSupabase Edge FunctionsRow Level SecurityPostgreSQL

Data modellingTailwind CSSshadcn/uiJWTAuthenticationIT securityDockerLinuxUbuntuVercel

CI/CDGitHub ActionsContinuous IntegrationVitestUnit TestingCursor AIModel Context Protocol (MCP)

Agentic CodingMVP developmentClean CodeSOLIDClean ArchitectureDomain-Driven Design

Digital transformationSoftware developmentInterface developmentAPI integrationThird-party system integration

IoTConsumer Electronics

LEGALTECH / NOTARY

01/2025 – 05/2025 · 5 MONTHS

05

DiNo — Digital Notary OfficeRole: Frontend Engineer Client: [LeXtorByte UG](#) ↗

↗ Testimonial



ROLE Frontend Engineer	TEAM 2–4 people	LOCATION Oranienburg, Germany · Remote
LANGUAGE German	WEBSITE digitales-notariat.de ↗	METHODOLOGY Scrumban

RESPONSIBILITIES

- Digital case management across several portals: structured data capture, digital case files and seamless internal filing.
- Status tracking in the client portal (partial contribution).
- User guidance for notarial processes that need explaining — every step comprehensible without legal training.
- Interfaces for the notary and client portals with **React** and **Material UI**.
- Initiative within the codebase: introduced reusable components, refactored existing code and fixed long-standing bugs — aimed at solutions that hold up over time.
- Unified form validation (**React Hook Form**) to speed up feature development across the whole team.
- Close coordination with design and backend in a **Scrumban** rhythm, turning new requirements into usable interfaces quickly.

DINO — DIGITAL NOTARY OFFICE — CONTINUED

RESULTS

- According to the provider, up to **70 %** fewer phone queries from clients.
- According to the provider, invoices are paid up to 30 days earlier.
- According to the provider, noticeably less administrative effort thanks to end-to-end digital processes.
- A consistent, modern UI across the client and notary portals, strengthening client trust and the notary office's efficiency alike.

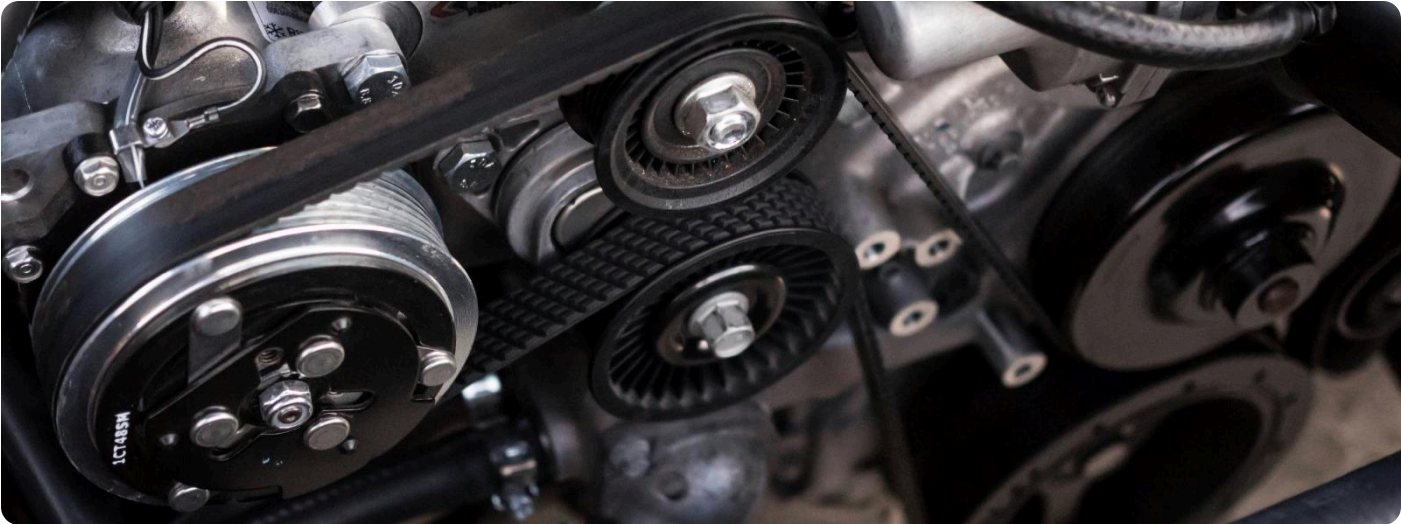
TECHNOLOGIES

TypeScript JavaScript Python SQL HTML CSS JSON React Flask REST API OpenAPI DTOs
 MariaDB MySQL Redux React Router React Hook Form Material UI (MUI) Component libraries
 Single Page Application (SPA) UI/UX Design Usability User guidance Frontend development
 Backend development Web development Client portal Document management Digital archiving
 Process digitalisation Digital transformation JWT OAuth 2.0 Role-Based Access Control (RBAC) GDPR Docker
 Nginx Linux Ubuntu GitLab GitLab CI Slack Confluence Clean Architecture Domain-Driven Design
 Code reviews Requirements Engineering Agile methods Software development Refactoring
 Legacy modernisation Requirements analysis LegalTech Notary services Scrumban

RETAIL & AUTOMOTIVE TRADE
01/2024 – 12/2024 · 12 MONTHS · ONGOING MAINTENANCE (SIDE PROJECT)

06 **AccountingOS — GoBD-Compliant Accounting & Warehouse System**

Role: Sole Developer · Full-Stack & AI Client: HD Autoservice [↗ Testimonial](#)



ROLE Sole Developer · Full-Stack & AI	TEAM Solo · client project	LOCATION Salzgitter, Germany · Remote
LANGUAGE German	VISIBILITY Internal system	METHODOLOGY Self-directed

RESPONSIBILITIES

- **Accounting** module: invoices, corrective invoices, PDF export, customer management, order import with **GoBD**-compliant record locking.
- Coupling of **warehouse management** and invoicing (automatic stock deduction, minimum-stock notifications).
- A resilient data synchronisation service ("pipe service") for supplier catalogues — up to 20 m CSV rows over TLS with a queue, caching, retry logic, validation and deduplication.
- Recursive resolution of foreign-key dependencies in the input dialogs, for consistent master data maintenance.
- Customer and **master data management** with **role-based access control** (admin, staff, tax adviser) and authentication via **better-auth**; type-safe **GraphQL** and REST interfaces (**NestJS**, **MikroORM**, **DTOs**).
- Technical advice for the client on approaches, architecture and process optimisation.
- Self-hosted delivery on **Linux** servers (**Docker**) with a **CI/CD** pipeline (**GitHub Actions**) and automated quality checks before every commit (Husky).
- Sole responsibility for the architecture (**clean architecture**), deployment and ongoing support (side project): maintenance, security updates and further development.

ACCOUNTINGOS — GOBD-COMPLIANT ACCOUNTING & WAREHOUSE SYSTEM — CONTINUED

RESULTS

- According to the client, up to 40 hours less administrative work per month — roughly 480 hours a year.
- According to the client, routine processes take up to **50 %** less time.
- End-to-end digital, **GoBD**-compliant workflows with warehouse and **accounting** directly coupled.
- In continuous operation since go-live: delivery, security updates and further development keep running automatically, without the client having to maintain an IT function for it.
- Daily automatic synchronisation of every supplier's car parts catalogue with **Accounting** OS — stock and prices are in place without a manual import.

TECHNOLOGIES

TypeScript JavaScript SQL HTML CSS JSON YAML Node.js NestJS Fastify Dependency Injection
 Next.js React Server-Side Rendering (SSR) TanStack Query Tailwind CSS Storybook REST API GraphQL
 DTOs Type safety Backend development Frontend development Web development Data synchronisation
 CSV processing ETL Message Queues Caching Idempotency Resilience Patterns MariaDB PostgreSQL
 Redis MikroORM TypeORM Data modelling Inventory management Warehouse management Accounting
 Invoicing Order management Master data management PDF generation ERP GoBD Audit-proof records
 Process digitalisation Digital transformation better-auth JWT Role-Based Access Control (RBAC) SSL/TLS
 IT security Docker Docker Compose Digital Ocean Self-Hosting Linux Ubuntu Monitoring Observability
 CI/CD Continuous Integration GitHub GitHub Actions Husky Miro Unit Testing Test automation
 Claude Code Cursor AI Model Context Protocol (MCP) Agentic Coding Agentic pull request workflows Monorepo
 Nx Microservices Domain-Driven Design Scalable architecture Clean Code SOLID Clean Architecture
 Requirements Engineering IT consulting Software development Product ownership Entrepreneurship
 Interface development API integration Requirements analysis

LOGISTICS & TRANSPORT

02/2021 – 05/2023 · 2 YEARS 4 MONTHS · ONGOING MAINTENANCE (SIDE PROJECT)

07

Ladetrans — Truck Route Management for a Humanitarian Aid Organisation

Role: Sole Developer · Full-Stack & AI Client: [CDH Stephanus](https://cdh-stephanus.org) ↗

ROLE Sole Developer · Full-Stack & AI	TEAM Solo · client project	LOCATION Bremen, Germany · Remote
LANGUAGE German	WEBSITE cdh-stephanus.org ↗	METHODOLOGY Self-directed

RESPONSIBILITIES

- Loading lists per trip: goods to be shipped, recipient addresses and drivers assembled in a single case.
- Every trip as one continuous case — create, edit, close and archive.
- Automated generation of the route documents from the captured data, among other things for border checks.
- Audit-proof archiving of completed trips including the proofs uploaded at the end of each trip.
- Sole responsibility for the backend architecture (**NestJS**, **clean architecture**), **role-based access control** (**better-auth**) and the self-hosted deployment including ongoing maintenance (side project).
- Automated delivery of the self-hosted environment: **CI/CD** pipeline (**GitHub Actions**), containerised build (**Docker**) behind **Nginx** as a reverse proxy, quality checks before every commit (Husky).

LADETRANS — TRUCK ROUTE MANAGEMENT FOR A HUMANITARIAN AID ORGANISATION — CONTINUED

RESULTS

- According to the client, automated document processes save up to 1,000 hours per employee per year in trip handling — the route documents, previously assembled by hand for every trip, have since been generated from the captured data.
- According to the client, up to **30 %** higher **logistics** efficiency thanks to end-to-end management of the shipping trips.
- Markedly fewer transcription errors; every trip is archived audit-proof together with its documents and proofs.
- In continuous use for the aid transports since 2021 — self-hosted, with automated delivery and ongoing maintenance from one source.

TECHNOLOGIES

TypeScript JavaScript SQL HTML CSS XML JSON YAML Node.js NestJS Fastify
 Dependency Injection Next.js React Server-Side Rendering (SSR) React Router Single Page Application (SPA)
 Tailwind CSS Storybook REST API GraphQL Backend development Frontend development Web development
 PostgreSQL Redis Caching MikroORM TypeORM Data modelling Logistics Route planning
 Order management Document management PDF generation DOCX generation Digital archiving
 Process digitalisation Digital transformation Platform migration System integration better-auth JWT
 Authentication Role-Based Access Control (RBAC) IT security Docker Docker Compose Digital Ocean
 Self-Hosting Linux Ubuntu Nginx Monitoring Observability CI/CD Continuous Integration
 GitHub Actions Husky Unit Testing Test automation Claude Code Cursor AI Model Context Protocol (MCP)
 Agentic Coding Agentic pull request workflows Monorepo Nx Slack Microservices Domain-Driven Design
 Scalable architecture Clean Code SOLID Clean Architecture Requirements Engineering IT consulting
 Software development Product ownership Interface development Requirements analysis

EDTECH / E-LEARNING

05/2020 – 02/2023 · 2 YEARS 10 MONTHS

08

XU Navigator — Enterprise Learning PlatformRole: Full-Stack Engineer Client: [XU Group](#) ↗[↗ Testimonial](#)

ROLE Full-Stack Engineer	TEAM 5–10 people	LOCATION Bremen, Germany · Remote
LANGUAGE German & English	WEBSITE xu.de ↗	METHODOLOGY Scrum

RESPONSIBILITIES

- Course content (videos, podcasts, multiple choice, articles) that each tenant fills dynamically, served by clearly bounded services.
- Learning progress tracking including automatic certificate generation.
- Expert and community forums for exchange between participants and specialists.
- Full-stack development across all platform modules within an agile **Scrum** team — from course management to the certificate engine, in frontend and backend alike.
- Scalable backend services (**NestJS**, **GraphQL**) with clearly bounded domains in a microservice architecture.
- The learning platform's frontend in **Angular** (**Angular Material**, **NgRx**), delivered through **Azure DevOps** pipelines.

XU NAVIGATOR — ENTERPRISE LEARNING PLATFORM — CONTINUED

RESULTS

- A scalable microservice platform running in production at well-known corporations (VW, ThyssenKrupp and others).
- Tenant-specific course content: every corporation runs its own training on the same platform.
- Certificates are issued without manual rework, and learning progress can be evaluated per tenant.
- Close, cross-functional collaboration with product and design over two and a half years — from the first build-out to production use at several major clients.

TECHNOLOGIES

TypeScript JavaScript HTML CSS SCSS XML JSON YAML Angular NgRx RxJS
 Angular Material Component libraries Figma UI implementation Node.js NestJS Express.js MEAN
 Dependency Injection REST API GraphQL Frontend development Backend development Web development
 MongoDB Mongoose NoSQL TypeORM Data modelling Microsoft Azure Azure DevOps Azure Pipelines
 Azure App Service Azure Functions Azure Blob Storage Docker CI/CD Continuous Integration
 Microsoft Teams JWT OAuth 2.0 Single Sign-On (SSO) Microsoft Entra ID (formerly Azure Active Directory)
 System integration Learning Management System (LMS) Video streaming Multi-Tenancy SaaS
 Platform development Monorepo Microservices Domain-Driven Design Scalable architecture
 Event-Driven Architecture Clean Architecture Agile methods Digital transformation Software development
 Interface development EdTech E-Learning Scrum

HEALTHTECH / MEDICAL TECHNOLOGY

01/2019 – 05/2020 · 1 YEAR 5 MONTHS

09

Medical Device Management System

Role: Full-Stack Engineer **Client:** Hospital group

ROLE Full-Stack Engineer	TEAM 2–4 people	LOCATION Germany · On site / Hybrid
LANGUAGE German	VISIBILITY Hospital group (anonymised)	METHODOLOGY Scrumban

RESPONSIBILITIES

- Device and **master data management** (hospital, location, device specifications).
- Detailed maintenance history for every device (when work was done and what kind) plus device-related document management.
- Fault reporting by nursing staff as the entry point into the medical engineering maintenance process.
- Multi-tenant filtering by hospital, plus integration of the Microsoft sign-in via **O365 Identity**.
- **REST API** with **ASP.NET Core (Swagger/OpenAPI)**; data modelling with **Entity Framework Core** (code-first) on **MSSQL**.

MEDICAL DEVICE MANAGEMENT SYSTEM — CONTINUED

RESULTS

- Multi-tenant device management for several hospitals — master data, maintenance history and document management per device in one place.
- Fault reports from nursing staff feed straight into the medical engineering maintenance process.
- A cleanly layered .NET architecture: code-first data model (**Entity Framework Core**, **MSSQL**), documented **REST API** (**Swagger/OpenAPI**) and authentication via **O365 Identity**.
- More than a year contributing within a small, closely aligned team on a safety-critical application in a hospital setting — with a focus on rigour, traceability and stable delivery.

TECHNOLOGIES

C#

Object-oriented programming (OOP)

.NET Core

ASP.NET Core

ASP.NET Core Web API

Dependency Injection

OpenAPI

Swagger

SQL

Entity Framework Core

Data modelling

Microsoft SQL Server (MSSQL)

TypeScript

JavaScript

HTML

CSS

XML

JSON

React

Redux

React Router

Axios

Fluent UI

Component libraries

Single Page Application (SPA)

Backend development

Frontend development

Web development

JWT

O365 Identity

Microsoft Entra ID (formerly Azure Active Directory)

Role-Based Access Control (RBAC)

Multi-Tenancy

System integration

Master data management

Document management

Maintenance management

Process digitalisation

Digital transformation

Microsoft Azure

Azure SQL

Azure DevOps

Docker

CI/CD

Continuous Integration

Visual Studio

Visual Studio Code

Microsoft Teams

Clean Architecture

Domain-Driven Design

Agile methods

Software development

Interface development

Requirements analysis

HealthTech

Medical technology

Scrumban

Skills

Full overview of my technical skills

01 AI, LLM & AI ENGINEERING

LLM integration RAG (retrieval augmented generation) Prompt Engineering Embeddings Vector databases
pgvector Semantic search Document analysis OCR LangChain OpenAI API Azure OpenAI
Anthropic Claude API Hugging Face Ollama AI agents Agentic Software Engineering AI-assisted automation
AI Engineering GenAI Artificial intelligence GDPR-compliant AI architecture

02 AGENTIC CODING

Claude Code Cursor AI GitHub Copilot Code Rabbit ChatGPT Lovable v0 Claude Code Hooks
Slash Commands Agent Skills Subagents Multi-agent workflows Model Context Protocol (MCP)
Agentic Coding Context Engineering Spec-Driven Development Agentic pull request workflows
AI-assisted test generation

03 PROGRAMMING LANGUAGES

TypeScript JavaScript Python C# SQL Bash HTML CSS SCSS XML JSON YAML UML

04 BACKEND & FRAMEWORKS

Node.js NestJS Express.js Fastify FastAPI Flask ASP.NET Core ASP.NET Core Web API .NET Core
MERN MEAN Headless CMS Strapi Directus GraphQL REST API OpenAPI Swagger gRPC
WebSockets Webhooks Message Queues BullMQ RabbitMQ Backend development
Object-oriented programming (OOP)

Skills

Continued

05 FRONTEND

React Next.js Angular React Native Ionic Capacitor Cross-platform development iOS Android
 Redux NgRx RxJS TanStack Query React Hook Form React Router Axios Zod Tailwind CSS
 shadcn/ui Material UI (MUI) Angular Material Fluent UI Single Page Application (SPA)
 Progressive Web App (PWA) Server-Side Rendering (SSR) Internationalisation (i18n) Frontend development
 Web development

06 DESIGN & UX

Figma Wireframing Prototyping Design System Design Tokens Component libraries Storybook
 UI/UX Design UI implementation Responsive Design Mobile First Accessibility (WCAG) Usability
 User guidance Onboarding flows Information architecture UX Writing

07 DATABASES & DATA

PostgreSQL MySQL MariaDB Microsoft SQL Server (MSSQL) MongoDB Mongoose NoSQL
 Cloud Firestore SQLite Redis Prisma MikroORM TypeORM Drizzle ORM Entity Framework Core
 Data modelling Data architecture DTOs Data synchronisation ETL CSV processing Caching
 Idempotency

08 DEVOPS, CLOUD & INFRASTRUCTURE

Docker Docker Compose CI/CD GitHub Actions GitLab CI Azure Pipelines Jenkins Nginx Sentry
 Datadog Monitoring Observability Linux Ubuntu Self-Hosting Microsoft Azure Azure Functions
 Azure App Service Azure Blob Storage Azure SQL Google Cloud Platform (GCP) Amazon Web Services (AWS)
 Hetzner IONOS Digital Ocean Vercel Supabase Supabase Realtime Supabase Edge Functions Firebase
 Cloud Functions Cloud Storage Firebase Cloud Messaging (FCM) Firebase Hosting DevOps DevSecOps

Skills

Continued

09 NETWORKING & PROTOCOLS

TCP/IP IP networks DNS HTTP/HTTPS

10 AUTH & SECURITY

JWT OAuth 2.0 Single Sign-On (SSO) better-auth Firebase Authentication Firebase Security Rules
Authentication Role-Based Access Control (RBAC) Row Level Security SSL/TLS End-to-end encryption
O365 Identity Microsoft Entra ID (formerly Azure Active Directory) Secure by Design IT security

11 ARCHITECTURE & PRINCIPLES

Software architecture Clean Architecture SOLID Design Patterns Domain-Driven Design
Separation of Concerns Dependency Injection Microservices Event-Driven Architecture Monorepo Nx
Turborepo Scalable architecture System integration Interface development API integration
Third-party system integration Platform migration Middleware API-First Resilience Patterns Type safety
MVP development

12 WAYS OF WORKING & METHODOLOGY

Agile methods Scrum Kanban Scrumban Requirements Engineering Requirements analysis
Sprint Planning Backlog Refinement Retrospective Daily Stand-up

Skills

Continued

13 ENGINEERING PRACTICES

Clean Code Code reviews Refactoring Legacy modernisation Continuous Integration Continuous Delivery

14 CONSULTING & BUSINESS

IT consulting Software development Project management IT strategy Digital strategy Digital transformation
Process digitalisation Product ownership Entrepreneurship

15 DATA PROTECTION & COMPLIANCE

GDPR Privacy by Design Data minimisation Data processing agreement (DPA) EU AI Act AI Governance
GoBD Audit-proof records EU hosting Digital sovereignty German Accessibility Reinforcement Act (BFSG)
European Accessibility Act (EAA)

16 TESTING & QA

Jest Vitest Cypress Playwright React Testing Library Unit Testing Integration tests End-to-end tests
Postman API testing Test automation

Skills

Continued

17 TOOLING

Git GitHub GitLab Husky npm pnpm CLI (Command Line Interface) Visual Studio Code
 Visual Studio Jira Confluence Linear Trello Azure DevOps Miro Slack Microsoft Teams
 Google Calendar Google Workspace (Meet, APIs) Bluedot HubSpot Cal.com n8n Amplitude
 Product Analytics

18 DOMAINS & INDUSTRIES

LegalTech GovTech Law firm software Notary services Client portal Document management HealthTech
 Medical technology Maintenance management ERP Inventory management Warehouse management
 Invoicing Order management CRM Accounting Master data management Multi-Tenancy Stripe SEPA
 Payment processing Logistics Route planning PDF generation DOCX generation Digital archiving EdTech
 E-Learning Learning Management System (LMS) Video streaming Social Media IoT Device Pairing
 Consumer Electronics SaaS Platform development

19 COMMUNICATION & PRESENTATION

Professional communication Clear communication (German/English) Proactive communication Presentation skills
 Public Speaking Stakeholder communication Constructive feedback Technical documentation

20 TEAMWORK & COLLABORATION

Teamwork Fitting into existing teams Collegial collaboration Knowledge sharing Remote collaboration
 Asynchronous collaboration Cross-functional collaboration

Skills

Continued

21 RELIABILITY & WORK ETHIC

Reliability Sense of responsibility Self-directed working Fast response times Results-driven
Solution-oriented Willingness to learn

22 ROLES & PROFILE

Senior Full-Stack Engineer AI Engineer Software Engineer Software developer Full-stack developer
Frontend developer Backend developer Web developer Software architect Freelance Developer
Freelance software developer Technical advisor IT specialist for application development Remote Work
German (native) English B2 Business English Munich DACH